

What Does Digital Red Queen Select For?

A Compute-Efficient Study of Pairwise vs. Multiplayer Selection

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Abstract

Digital Red Queen (DRQ) evolves Core War programs through simultaneous multiplayer battles, while evaluating held-out generality through 1-v-1 matches. This manuscript compares the selection pressures induced by simultaneous multiplayer evaluation and continuous pairwise evaluation against the same historical opponents, and investigates the origin of their disagreements. We rerank the strongest candidates from several rounds and independent evolutionary roots using both objectives. The candidate selected by multiplayer evaluation is not necessarily the strongest pairwise candidate. More broadly, differences between native DRQ selection and pairwise rankings reflect sensitivity to placement seeds, sensitivity to warrior list order, and multiplayer effects that cannot be predicted from isolated performances. These interaction effects remain after seeds and order are controlled and, in some cases, emerge only when several opponents coexist. The two objectives therefore impose distinct selection pressures: replacing one with the other can change selection decisions and could therefore redirect the evolutionary process.

1 Introduction

AI-driven improvement systems progress through repeated cycles of generation, evaluation, and selection. Their evaluator does more than measure performance: it determines which discoveries are preserved and which directions are explored next. In adversarial environments, this distinction is especially important because performance against several simultaneous opponents may not be predictable from isolated encounters.

Digital Red Queen (DRQ) studies AI-driven program evolution through Core War, a game in which programs compete for control of a shared memory [1]. Within each round, an LLM generates mutations and MAP-Elites maintains a diverse candidate archive [2]. Candidates face an expanding historical panel composed of the initial warrior and the champions selected in previous rounds, and each new winner is added to that panel. Evolutionary selection is performed through simultaneous multiplayer battles, whereas held-out generality is evaluated through 1-v-1 matches against unseen human-written warriors. Because each selected champion shapes the environment of later rounds, a difference between these evaluation modes could redirect the evolutionary trajectory.

This manuscript compares the selection pressures induced by DRQ’s simultaneous multiplayer evaluation and by continuous pairwise evaluation against the same historical opponents. The two scores answer different questions. Pairwise evaluation asks whether a candidate is robust across

isolated matchups; simultaneous evaluation asks whether it remains effective when the same opponents coexist. We therefore investigate whether they induce the same selection pressure and, when they disagree, which properties of the evaluation protocol and multiplayer context produce the difference. The analysis first determines whether the two objectives rank the strongest candidates similarly and preserve the same winner, then separates sensitivity to placement seeds and warrior order from effects that emerge only when several opponents coexist.

2 Experimental setup

2.1 Generating the evolutionary histories

Our analyses are based on four independently generated DRQ histories, hereafter referred to as roots 0–3. The roots started respectively from the human-written warriors `fizzle.red`, `catcan.red`, `garlic.red`, and `wuss.red`. They used distinct language-model and simulator seeds but otherwise followed the same configuration, allowing the comparison to cover several evolutionary trajectories rather than a single run.

Each root contains four evolutionary rounds. During round q , a candidate is evaluated simultaneously with the initial human-written warrior and the champions selected at the end of rounds 1 through $q - 1$. The arena therefore grows with the history: it contains two warriors in round 1, three in round 2, four in round 3, and five in round 4. At the end of each round, the highest-fitness candidate becomes the new champion and is added to the opponent panel used in the following rounds. Every historical slot is retained, including when the same executable wins more than one round.

Within each round, MAP-Elites maintains an archive indexed by two behavioural descriptors: the total number of processes spawned by the program and the proportion of memory locations it accesses. The archive is warm-started with the initial warrior and the champions from previous rounds, and eight new programs are generated before the main search begins. The search then runs for 250 iterations. At each iteration, the language model either generates a new program with probability 0.1, or mutates a program sampled uniformly from the occupied cells of the archive. Each request produces one candidate, which is evaluated and inserted into the archive if it improves the corresponding cell.

All battles use the same Core War configuration as the original DRQ experiments: an 8,000-instruction core, a maximum of 80,000 cycles, at most 8,000 processes, a maximum warrior length of 100 instructions, and a minimum separation of 100 instructions between programs.

2.2 Native fitness and champion selection

DRQ does not reduce a battle to a binary victory or defeat. Instead, it assigns each warrior a continuous score that accounts for how long it remains alive and how many competitors survive alongside it. Consider a battle B containing N warriors, run with placement seed s for at most T cycles. Let the binary indicator A_t^i equal one when warrior i is alive at cycle t , and zero otherwise. Its fitness is

$$F_N(w_i; B, s) = \frac{N}{T} \sum_{t=1}^T \frac{A_t^i}{\sum_{j=1}^N A_t^j} \quad (1)$$

At each cycle, the available reward is divided equally among the surviving warriors. A program therefore benefits both from remaining alive and from eliminating competitors early enough to receive a larger share of the subsequent reward. The factor N/T gives a score of one to every warrior when all N participants survive until the cycle limit. The score is continuous, but it is also contextual: eliminating one warrior changes the reward subsequently received by every other survivor.

During evolution, a candidate h is placed in a fixed list order with the complete historical panel for the current round. Its native fitness is the mean of its scores over 20 placement seeds:

$$F_{\text{native}}(h) = \frac{1}{20} \sum_{s \in \mathcal{S}_{\text{native}}} F_N(h; B(h, \mathcal{H}), s) \quad (2)$$

Here, $\mathcal{H}$ contains the initial warrior and all previously selected champions. This stored mean determines insertion into the MAP-Elites archive and, ultimately, which candidate becomes the round champion.

Because the upper range of this fitness grows with the number of warriors, later experiments divide score differences by N when comparing margins across arena sizes. This normalization changes only their magnitude: within a fixed context, it cannot change the candidate ranking.

2.3 Computational workflow

Generating a DRQ history requires both repeated Core War simulations and access to a language model. We initially considered running the complete pipeline on the Jean Zay supercomputer using a locally hosted Qwen model, since its compute nodes cannot access the OpenAI API. Preliminary tests, however, produced substantially slower inference and less diverse and lower-quality programs than GPT-4.1 mini. The four reported roots were therefore generated with `gpt-4.1-mini-2025-04-14` at temperature 1.0 on a rented 16-thread OVHcloud CPU. This choice also kept the generation process closer to the original DRQ setup. The four roots were executed successively and required approximately 10.1 hours of cumulative wall-clock time.

Once the evolutionary histories had been generated, the remaining experiments required only the Core War simulator. The reranking, control, and mechanistic analyses were therefore run on Jean Zay, using 40 CPU cores per job. Independent jobs could be executed concurrently, and no additional language-model calls or program generation occurred during these post-hoc evaluations.

Across the take-home project, program generation required 5,036 API requests, comprising 20,671,243 input tokens and 1,299,211 output tokens. Of the input tokens, 18,691,456 were served from cache. The resulting API cost was USD 3.63: USD 1.43 for cached input, USD 0.60 for uncached input, and USD 1.60 for output.

3 Experiments

3.1 Does pairwise evaluation change rankings and selection?

Motivation

The first experiment addresses the most direct consequence of pairwise decomposition: does replacing the simultaneous arena with independent duels change which candidates are preferred? Any inversion in the ranking would show that the two objectives are not interchangeable. A disagreement at rank 1 would have a stronger consequence, since the selected winner is added to the historical panel and shapes the next evolutionary round. We therefore examine both the extent of reranking across the population and whether it changes the selection decision itself.

Protocol

We construct one comparison for each combination of four independent roots and evolutionary rounds 2–4, giving 12 root-rounds in total. In each case, we retain the 12 highest-fitness distinct valid candidates according to the stored native DRQ score. This restriction concentrates computation on the part of the population that can plausibly affect selection while leaving enough candidates to detect changes beyond rank 1. Round 1 is omitted because the candidate then faces only the initial warrior: the arena already contains two warriors, so simultaneous and pairwise evaluation have the same interaction structure. Each retained candidate is then reevaluated under both objectives against an identical historical panel and a common set of 24 newly sampled placement seeds. These seeds are distinct from those used during evolution.

The stored native score cannot serve directly as the multiplayer reference because it differs from the pairwise evaluation in two respects unrelated to interaction structure. Native DRQ used 20 placement seeds and one fixed list order, whereas our pairwise evaluation uses 24 fresh seeds and both possible duel orders. Comparing them directly would therefore mix the effect of decomposing the arena with seed resampling and positional effects. We instead rerun the complete simultaneous battle and average the candidate’s fitness over the $2N$ distinct forward and reversed cyclic orders:

$$M(h; \mathcal{H}) = \frac{1}{|\mathcal{S}| |\mathcal{O}_{\text{bal}}|} \sum_{s \in \mathcal{S}} \sum_{\pi \in \mathcal{O}_{\text{bal}}} F_N(h; \pi(h, \mathcal{H}), s) \quad (3)$$

Here, $\mathcal{H}$ is the historical panel, $N = |\mathcal{H}| + 1$, $\mathcal{S}$ is the set of fresh seeds, and $\mathcal{O}_{\text{bal}}$ is the balanced order set. This balanced score M preserves the defining feature of the native objective—all historical opponents coexist in the same arena—while matching the fresh placements and treating list position symmetrically. It therefore provides the appropriate multiplayer counterpart for isolating the effect of pairwise decomposition. Section 3.2 later quantifies the effects removed by these controls.

Using the same seeds and historical panel, the pairwise objective evaluates the candidate against each historical warrior in isolation. Every duel is run in both list orders, and every historical slot receives equal weight:

$$P(h; \mathcal{H}) = \frac{1}{|\mathcal{H}|} \sum_{w \in \mathcal{H}} \frac{1}{2|\mathcal{S}|} \sum_{s \in \mathcal{S}} [F_2(h; (h, w), s) + F_2(h; (w, h), s)] \quad (4)$$

The average is taken over historical slots rather than unique programs. When the same champion survives several rounds, several copies of it are present in the simultaneous history. Retaining those repeated slots in P keeps the two objectives matched on the curriculum they evaluate instead of silently changing the weight assigned to that champion.

Although P is computed from 1-v-1 battles, it should not be confused with the paper’s generality metric G . Generality is a held-out evaluation over 317 unseen human-written warriors and measures the fraction of them that the candidate defeats or ties:

$$G(h; \mathcal{U}) = \frac{1}{|\mathcal{U}|} \sum_{u \in \mathcal{U}} \mathbf{1}[\bar{F}_2(h, u) \geq \bar{F}_2(u, h)] \quad (5)$$

The barred $\bar{F}_2$ denotes two-player fitness averaged over placements. This binary statistic is well suited to measuring external robustness, but it discards information that is important during evolution: a narrow win and a dominant win both contribute one, while every loss contributes zero regardless of its severity. By contrast, P applies the same continuous Core War fitness used by DRQ to the existing historical panel. It distinguishes early elimination from survival until the cycle limit and can therefore impose a strong penalty on a severe matchup-specific failure. The definition of P is thus guided by its intended role, rather than chosen after observing the results: it is a continuous pairwise decomposition of the selection objective, whereas G remains a held-out endpoint. We do not assume that optimizing P would necessarily improve G .

After computing M and P , we compare their rankings using Kendall’s τ_b , the fraction of discordant candidate pairs, and an indicator of whether the top-ranked candidate set changes. The first two quantities describe the extent of reranking across the 12 candidates, while the last isolates the decision that would affect the next evolutionary round. We then inspect opponent-level duel scores to determine which matchups are responsible for the largest movements.

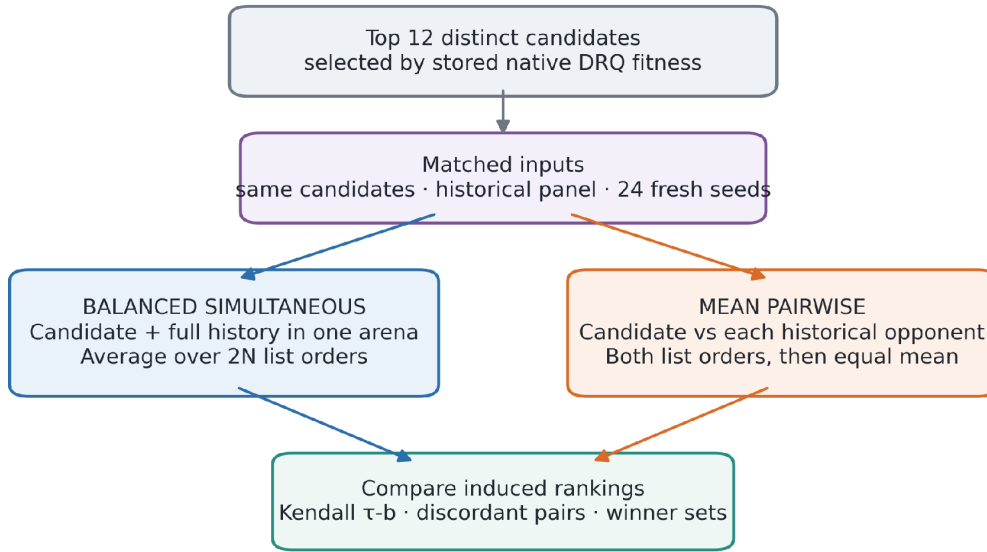
Results and analysis

Figure 2 presents the 12 root-round comparisons. Each cell reports global rank agreement, the fraction of candidate pairs whose relative order is reversed, and whether the winner set changes. Reading these quantities together distinguishes reranking elsewhere in the population from reranking that crosses the selection boundary.

Every root-round contains some reranking, with 161 of the 792 candidate pairs reversing order overall. In seven of the 12 comparisons, this disagreement reaches the top of the ranking and changes the winner set. One of these cases, root 0 at round 2, is only a numerical near-tie, but the other six changes across roots 1–3 are supported by clear continuous margins. Pairwise decomposition can therefore alter the candidate that would be selected, rather than merely rearranging secondary candidates.

The magnitude of this disagreement does not evolve monotonically with the number of rounds. Roots 0 and 1 begin with similar levels of agreement but then follow opposite trajectories. By round 4, root 0 reaches the highest τ_b in the matrix (0.839), whereas root 1 reaches the only

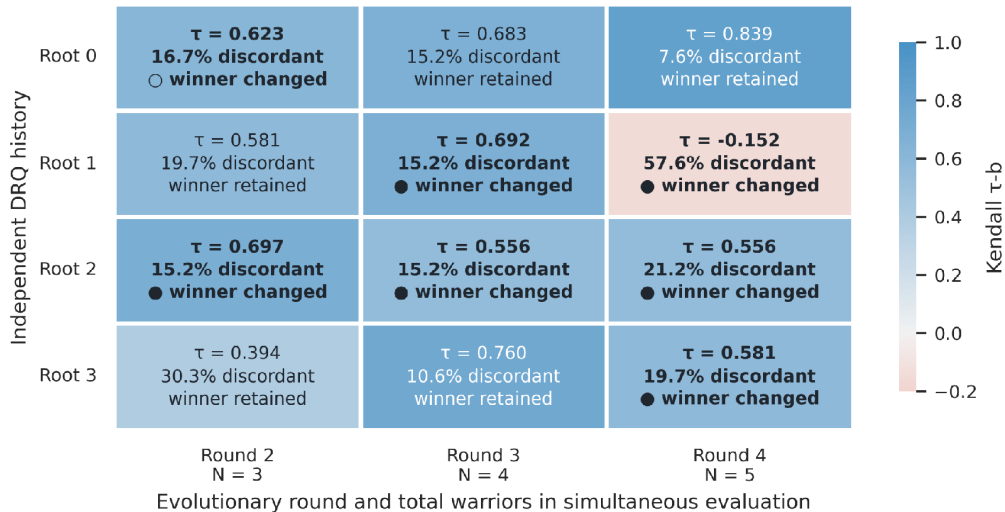
Matched reranking isolates simultaneous context from pairwise decomposition



Follow-up controls: native → fresh fixed order → balanced 2N orders → exhaustive N! orders; context depth N = 3, 4, 5

Figure 1. Matched reranking protocol. The candidate pool, historical panel, and 24 fresh placement seeds are held fixed while the interaction structure changes. Follow-up controls isolate seed, order, and context-depth effects.

Balanced simultaneous and mean pairwise rankings are not interchangeable



Across all 12 root-rounds: mean $\tau_b = 0.567$; 161/792 candidate pairs discordant (20.3%); winner set changed in 7/12.
 ● clear winner-set change ○ formal change associated with a numerical near-tie (root 0, round 2)

Figure 2. Agreement between balanced simultaneous and mean pairwise rankings. Each cell reports Kendall's τ_b for one root and round. Markers indicate whether the top-ranked candidate set changed. The open marker denotes the numerical near-tie in root 0, round 2.

negative value (-0.152). The other histories show why this cannot be reduced to a distinction between one stable run and one pathological run. Root 2 keeps a positive rank correlation at every round while selecting a different winner each time. Root 3 follows a non-monotonic trajectory: its relatively low agreement at round 2 leaves the winner unchanged, yet its higher agreement at round 4 produces a different winner. The amount of global reranking and its effect at rank 1 are therefore related but distinct, and neither is explained by round number or historical-panel size alone.

Figure 3 shows what these different regimes look like at round 4. In root 0, most candidates keep similar positions and the winner is preserved. Root 1 instead undergoes an almost population-wide reordering. Root 2 lies between these two cases: the rankings remain positively associated overall, yet their optima are meaningfully different. Replacing its simultaneous winner with the pairwise winner costs 0.193 normalized units under simultaneous evaluation, while making the opposite choice costs 0.082 under pairwise evaluation. The winner reversal is therefore not an arbitrary consequence of a tie at rank 1.

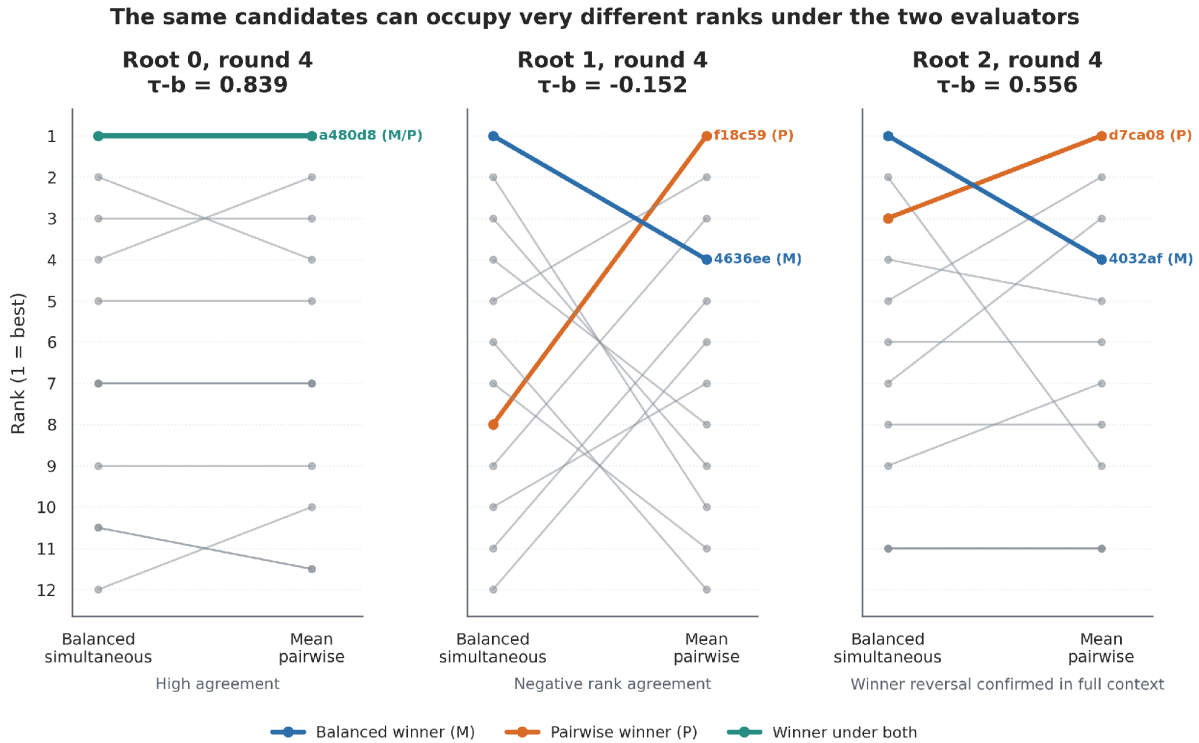


Figure 3. Representative rank movements in round 4. Lines connect each candidate’s balanced-simultaneous and mean-pairwise ranks; rank 1 is best. Blue and orange identify the winners under the two objectives.

These rank plots show where the objectives disagree, but they do not explain why. Because P is an average of continuous opponent-specific scores, its components can be inspected directly. Figure 4A relates each candidate’s change in rank to its worst-matchup deficit, defined as its lowest duel score after subtracting the median score of the 12 candidates against that opponent. Figure 4B then focuses on the six clear winner reversals and shows, for every historical slot, the fitness difference between the pairwise and simultaneous winners.

Large pairwise rerankings expose matchup-specific vulnerabilities

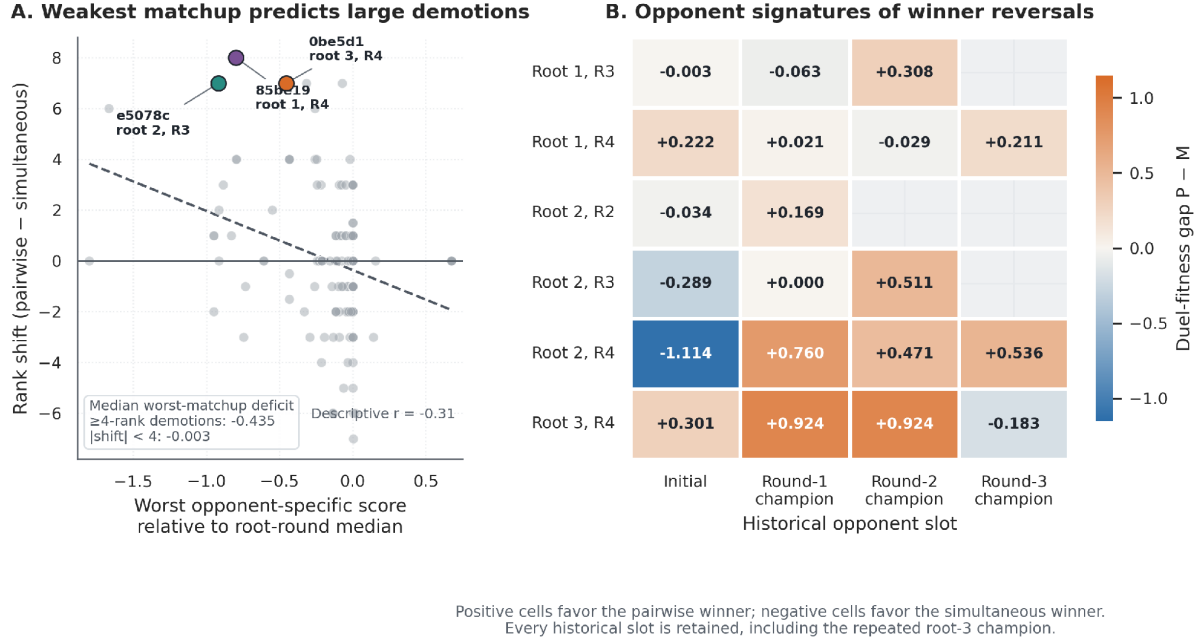


Figure 4. Opponent-level sources of pairwise reranking. Panel A relates each candidate’s pairwise rank shift to its worst opponent-specific deficit; highlighted points are large demotions discussed in the text. Panel B reports $\bar{F}_2$ (pairwise winner) minus $\bar{F}_2$ (simultaneous winner) against every historical slot for the six clear winner changes. Positive cells favor the pairwise winner. Empty cells correspond to opponents not yet present, and the repeated root-3 champion is retained twice.

Figure 4A indicates that the largest demotions are often driven by a concentrated weakness rather than a uniform decline against the entire history. The association is modest across all 144 candidate-round records (descriptive $r = -0.31$), but becomes much clearer in the tail. The 15 candidates demoted by at least four ranks have a median worst-matchup deficit of -0.435, compared with -0.003 for candidates moving fewer than four ranks in either direction. Candidate e5078c in root 2, round 3 illustrates the mechanism particularly clearly. It ranks fifth under simultaneous evaluation and obtains scores of 1.000 and 1.118 against two opponents, yet scores only 0.084 against *garlic.red*; this isolated failure pushes it to rank 12 under P . Candidate 85be19 in root 1, round 4 exhibits the same pattern less dramatically. Although it ranks second in multiplayer and performs best against the latest champion, its score against the round-1 champion lies 0.797 below the candidate median for that matchup, contributing to a fall to pairwise rank 10. Pairwise evaluation is therefore not uniformly downgrading these candidates. It is exposing severe weaknesses that their simultaneous performance can conceal.

Panel B shows that the same mechanism can change the winner in more than one way. At round 3, the decision can hinge on a single recent opponent. In root 1, the pairwise winner is marginally worse against the first two historical slots (-0.003 and -0.063) but gains 0.308 against the round-2 champion. Root 2 follows the same pattern: an advantage of 0.511 against the latest champion outweighs a deficit of 0.289 against the initial warrior. By round 4, the disagreement can extend across a larger part of the history. The pairwise winner in root 1 is better against three of four opponents; in root 2, it trails the simultaneous winner by 1.114 against the initial warrior but leads by 0.471–0.760 against each evolved champion. These preferences are determined by the size of the margins, not by a simple vote across matchups. In 96 of the 161 discordant candidate

pairs, the candidate preferred by P is better in at most half of the historical slots. A few decisive advantages, or one severe failure, can therefore outweigh several narrow results - precisely the information that a binary win-rate metric would discard.

Root 3 adds a further complication: the evolutionary history can amplify a vulnerability through multiplicity. The same champion survives rounds 1 and 2 and consequently occupies two historical slots. At round 4, the simultaneous winner scores 0.780 against this program, whereas each of the tied pairwise winners scores 1.704. Treating the champion as two slots is necessary for a matched comparison because two copies also coexist in the simultaneous battle. Under P , however, the corresponding isolated duel contributes twice to the average, helping move the simultaneous winner from rank 1 to rank 8. The effect of a history therefore depends not only on how many opponents it contains, but also on which opponents recur.

Together, these observations establish the central result of the first experiment: pairwise and simultaneous evaluation are related but not interchangeable, and their disagreement can reach the selection boundary. The opponent-level decomposition also clarifies what P rewards. Because it averages continuous duel margins, P exposes concentrated matchup-specific vulnerabilities that M can conceal, and a single severe failure can outweigh several narrow advantages. Repeated historical slots can amplify the same effect. These mechanisms explain many downward rank shifts under P , but they do not explain why some candidates preferred by P in isolation lose their advantage when the opponents coexist. Two complementary analyses follow. Section 3.2 tests whether balancing placement seeds and list order was necessary to isolate the matched comparison; Section 3.3 returns to the residual disagreement between M and P and reconstructs the multiplayer context in which it emerges.

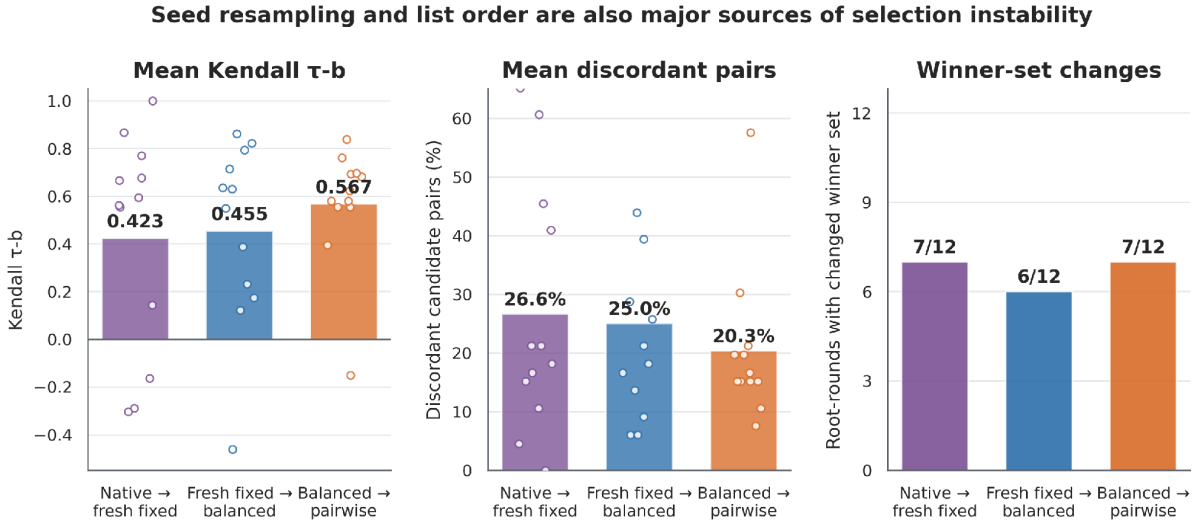
3.2 Seed and order effects do not explain away the matched disagreement

Section 3.1 compared M with P under matched conditions in order to isolate one change: replacing a simultaneous arena with independent 1-v-1 battles. The candidate pool and historical panel were identical, both scores used the same 24 fresh placement seeds, and list position was treated symmetrically. This construction differs from native DRQ, which uses 20 placement seeds and one fixed list order. The present control tests whether those differences matter. Starting from the stored native fitness, we first change only the placement sample by recomputing multiplayer fitness on the 24 fresh seeds while retaining the native fixed order. We then keep those seeds fixed and balance list order. If either step changes the ranking, comparing native fitness directly with P would confound multiplayer decomposition with seed or position effects, validating M as the appropriate reference for Section 3.1.

The first transition changed only the placement sample. Nevertheless, mean τ_b fell to 0.423, 26.6% of candidate pairs reversed, and the winner changed in 7 of 12 comparisons. In this short-run regime, a stored DRQ ranking is therefore not a stable property of the programs alone; it also reflects the 20 placements on which it happened to be measured.

The second transition kept the 24 fresh seeds but averaged the forward and reversed cyclic orders. It produced mean $\tau_b = 0.455$, 25.0% discordant pairs, and 6 of 12 winner changes. Order sensitivity was thus comparable in magnitude to seed resampling. This explains part of the gap between stored native fitness and the final pairwise ranking, and justifies using a symmetrized multiplayer comparator for the decomposability question.

Only the final transition changed the interaction structure. Balanced simultaneous and pairwise evaluation used the same candidates, historical panel, and fresh seeds, while controlling list position within their respective contexts. Agreement increased to $\tau_b = 0.567$, but 20.3% of pairs and 7 of 12 winner sets still changed. Seed and order effects are therefore major sources of selection instability, not minor implementation details. At the same time, they cannot account for the residual matched disagreement: it remains after both have been controlled by design. The three contrasts are diagnostic rather than additive, but together they narrow the explanation from generic simulator noise to the structure of the evaluation context.



Bars are descriptive means across 12 root-rounds; points show individual root-rounds. Rounds within a root are repeated measures.

Figure 5. Sequential controls for seed, order, and interaction structure. Bars summarize agreement between adjacent evaluator definitions; points show individual root-rounds. The contrasts are diagnostic rather than an additive variance decomposition.

This control analysis therefore validates balanced M as the reference used in Section 3.1: seed resampling and order balancing both change rankings enough that a direct comparison between native fitness and P would not isolate multiplayer decomposition. At the same time, these controls do not explain away the remaining disagreement between M and P , because it persists after the candidates, opponents, seeds, and treatment of list position have been matched. Section 3.3 now returns to this residual difference and examines how it appears as the simultaneous context is progressively reconstructed.

3.3 Tracing preference reversals through multiplayer context depth

Motivation

Section 3.1 explains one side of the disagreement between M and P : pairwise evaluation exposes severe opponent-specific weaknesses that a candidate can survive in the simultaneous arena. It does not explain the converse observation. If a candidate loses to several historical opponents in isolation, what allows the same candidate to recover once those opponents share an arena?

The defining structural difference between the objectives provides a possible answer. P isolates each historical opponent before averaging the resulting scores, whereas M allows the opponents to interact with one another and jointly alter survival times, memory occupation, and reward sharing.

We therefore reconstructed the simultaneous context progressively. The aim was not simply to confirm another ranking change, but to determine when it appears. If one additional opponent were sufficient, the disagreement could be attributed to a relatively simple third-party interaction. If the preference changed only in richer contexts, it would instead provide evidence that the multiplayer objective contains a genuinely higher-order component.

Protocol

We first studied the round-4 winner reversals in roots 1 and 2. Candidate A denotes the balanced-simultaneous winner and candidate B the pairwise winner. Root 0 was subsequently added as a control. Because M and P select the same winner there, A was compared with the pairwise runner-up, giving a case in which the preference should remain stable. The mechanistic experiments were not run for root 3.

For every context, we follow the normalized margin

$$\Delta_N(A, B) = \frac{\bar{F}_N(A) - \bar{F}_N(B)}{N} \quad (6)$$

where N is the total number of warriors and $\bar{F}_N$ is the candidate's fitness averaged over the evaluated seeds and list orders. A and B are evaluated separately against the same opponents. Positive values favor the simultaneous winner A , negative values favor B , and a change of sign marks a preference reversal. The normalization makes margins comparable across context sizes without changing the preference within any context.

The pairwise margin provides the starting point. Three-warrior evaluation then places each candidate with every pair drawn from the four historical opponents, producing six contexts. Four-warrior evaluation uses every historical triple, producing four contexts, before five-warrior evaluation restores the complete round-4 panel. At each depth, we evaluate every possible list order: $3! = 6$, $4! = 24$, and $5! = 120$ orders respectively. Every order uses the common set of 24 fresh seed values from Section 3.1, not the seeds used during evolution.

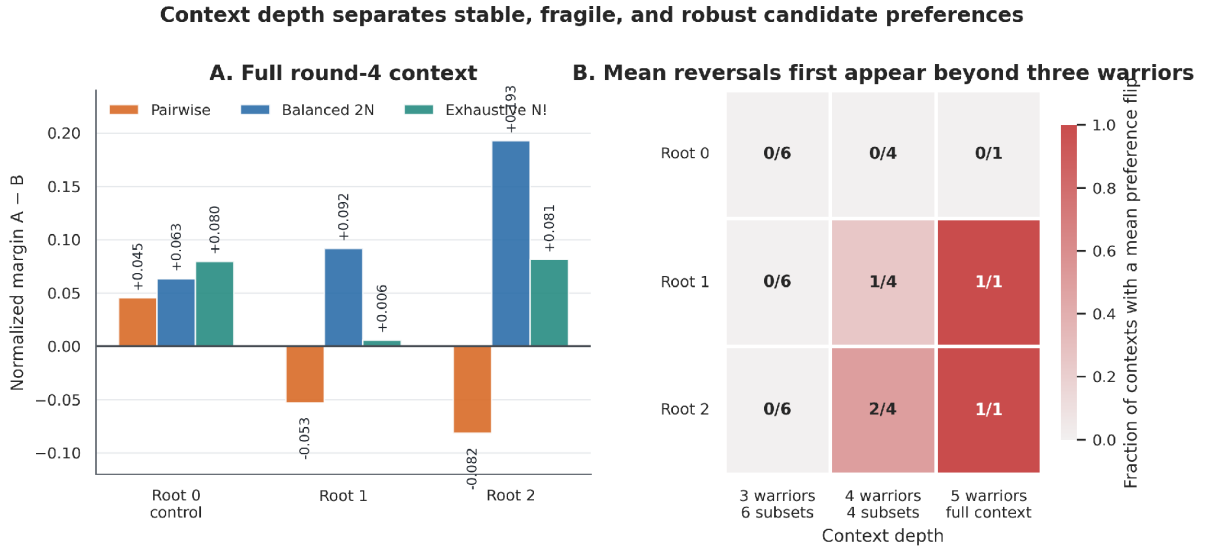
The final step also audits the order approximation used by M . Population-wide reranking averages the $2N = 10$ forward and reversed cyclic orders, whereas the five-warrior audit covers all 120 permutations. Comparing the two tells us whether the detected sign survives exhaustive order averaging and whether the ten-order margin accurately represents its strength.

Results and analysis

Figure 6 summarizes how the preference evolves with context depth. No three-warrior context reverses the mean pairwise preference in any root. Some individual orders already favor the other candidate in roots 1 and 2, but their effects cancel when the six orders of a context are

averaged. For the candidate pairs studied here, one additional opponent is therefore insufficient to reproduce the disagreement.

The first mean reversals appear when three historical opponents coexist with the candidate. None of the four such contexts reverses the preference in root 0, compared with one in root 1 and two in root 2. Restoring the fourth opponent then preserves both winner reversals, but exhaustive order averaging separates a fragile case from a robust one. Root 1 moves from a pairwise margin of -0.053 to only $+0.006$ across all 120 orders, whereas root 2 moves from -0.082 to $+0.081$. Root 0 remains positive throughout. Interaction depth can therefore change the selected candidate, but the depth-level counts alone conceal why the two reversals have such different strengths.



Candidate A is the balanced-simultaneous winner; candidate B is the pairwise winner (roots 1-2) or pairwise runner-up control (root 0).

Figure 6. Candidate preference as interaction depth increases. Candidate *A* is the balanced-simultaneous winner; candidate *B* is the pairwise winner in roots 1 and 2 and the pairwise runner-up in the root-0 control. Panel A compares the normalized pairwise, ten-order balanced, and exhaustive full-context margins. Panel B reports how many contexts at each depth reverse the corresponding pairwise preference.

Figure 7 resolves this difference at the level of particular opponent sets and list orientations.

The clearest composition effect occurs in root 2. *A* beats the initial warrior **garlic.red** by $+0.557$ in normalized pairwise margin, but loses to the three evolved champions by -0.380 , -0.235 , and -0.268 respectively. *P* averages these four isolated matchups and consequently favors *B*. Simultaneous evaluation divides the same panel differently: every tested context containing garlic favors *A*, while every context without garlic favors *B* (Figure 7A). This separation holds for both the three- and four-warrior subsets.

The four-warrior ablations make garlic’s role especially clear. Removing it changes the simultaneous margin from $+0.081$ in the complete context to -0.245 , whereas removing champions 1 or 3 leaves the preference close to its full-context value. Removing champion 2 instead increases *A*’s advantage to $+0.247$, showing that the evolved opponents do not contribute uniformly: champion 2 partly counteracts the reversal. Garlic is therefore necessary for *A* to remain preferred in the tested multiplayer contexts, but it is not a third party that independently flips a duel. Subsets containing garlic already favor *A* under their pairwise baseline. The non-decomposable

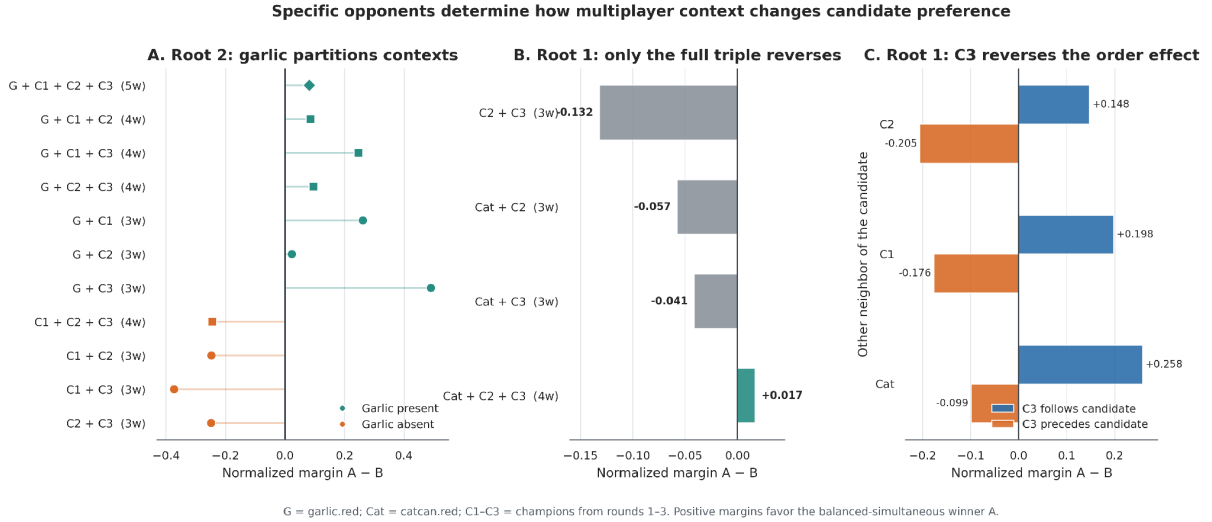


Figure 7. Opponent-specific mechanisms behind the round-4 preferences. Panel A reports every simultaneous context tested in root 2 and shows that the sign is partitioned by the presence of **garlic.red**. Panel B isolates the only reversing four-warrior context in root 1: none of its three constituent opponent pairs favors A, but their union does. Panel C groups the exhaustive root-1 orders by the cyclic neighbors of the candidate and shows that placing champion 3 immediately before or after it reverses the mean preference. *G* denotes **garlic.red**, *Cat* denotes **catcan.red**, and *C*₁-*C*₃ denote the champions from rounds 1-3.

result appears only when several evolved opponents make *P* favor *B* while their simultaneous coexistence remains favorable to *A*.

The change in root 2 is also better described as a relative collapse of *B* than as an absolute improvement of *A*. After normalization, *A* falls from 0.601 pairwise to 0.458 in the exhaustive context, while *B* falls from 0.683 to 0.377. Both candidates perform worse as the arena becomes crowded, but *B* loses 0.163 more. The interaction between garlic and champion 3 provides the strongest small-context example: their simultaneous margin is +0.490, compared with +0.145 for the corresponding pairwise average. Most of this shift comes from *B*, whose normalized score decreases by 0.287, while *A* gains only 0.059. The multiplayer winner is not simply being rewarded for stronger isolated matchups; it is more resistant to the environment created when these opponents coexist.

Root 1 exhibits a different mechanism. Its pairwise winner gains most of its advantage against **catcan.red** and champion 3, but champion 3 also makes that advantage sharply dependent on cyclic list orientation. In the three-warrior context containing catcan and champion 3, the orientation candidate → Cat → *C*₃ favors *B* by 0.323 on average, whereas reversing the cycle to candidate → *C*₃ → Cat favors *A* by 0.241. *A*'s score changes only moderately between these orientations; *B*'s normalized score falls from approximately 0.75 to 0.24.

The same directional pattern survives in the complete context. Whenever *C*₃ immediately follows the candidate, the mean *A*-minus-*B* margin is positive, ranging from +0.148 to +0.258 depending on the candidate's other neighbor. Whenever *C*₃ immediately precedes the candidate, it is negative, ranging from -0.099 to -0.205 (Figure 7C). By comparison, grouping the 120 orders by the candidate's absolute list position yields means between only -0.015 and +0.037. Root 1 is therefore not dominated by one generally advantageous slot. Its near-zero exhaustive margin results from two opposed orientation regimes centered on champion 3.

The composition responsible for the root-1 mean reversal is itself non-additive. Its three opponent pairs—catcan with C_2 , catcan with C_3 , and C_2 with C_3 —all favor B in simultaneous evaluation, with margins of -0.057, -0.041, and -0.132. When the three opponents are combined, the margin becomes +0.017 (Figure 7B). No constituent pair therefore predicts the sign of their union. This is direct evidence of a higher-order effect, but not a strong selection preference: exactly half of the 24 orders favor each candidate, and adding the remaining champion reduces the exhaustive margin to +0.006.

Root 0 provides the complementary control. A is better than or effectively tied with B against every opponent in isolation, and every three- and four-warrior context retains that preference. All 120 full-context orders also favor A , with margins between +0.023 and +0.137. The margin can nevertheless change substantially with composition; for example, the context containing champions 2 and 3 increases it from +0.060 under the corresponding pairwise average to +0.145 simultaneously. Higher-order effects can therefore modify the strength of a preference without becoming large enough to alter selection.

The exhaustive audit also qualifies the ten-order approximation without overturning it. For the three candidate pairs audited here, the balanced subset preserves the exhaustive preference sign, which supports its use as a practical population-wide comparator in Section 3.1. It is less accurate for effect size: the root-1 margin falls from +0.092 balanced to +0.006 exhaustive, and the root-2 margin from +0.193 to +0.081. The balanced score can therefore identify candidate-level disagreement at manageable cost, while exhaustive permutations remain necessary before interpreting an individual reversal as robust.

Taken together, these audits show why the remaining disagreement cannot be attributed to one universal multiplayer mechanism. Depending on the evolutionary history, it can be driven by opponent composition, amplified by local orientation, or remain too weak to alter selection. Because the analysis follows one candidate pair in only three roots, it identifies possible mechanisms rather than estimating how frequently each occurs. The present experiments establish evaluator-dependent selection differences under matched finite samples, but do not quantify the stability of every winner reversal across independent placement samples or exhaustive order averaging. The final section considers what this distinction implies for evaluator choice and for future evolutionary experiments.

4 Conclusion

Simultaneous multiplayer and continuous pairwise evaluation do not impose the same selection pressure. Their differences partly reflect sensitivity to placement seeds and warrior order, but these factors do not explain the entire disagreement. Some preferences arise only when several opponents interact in the same arena and cannot be recovered from their isolated 1-v-1 performances. Pairwise evaluation is therefore a coherent alternative objective, particularly because it exposes severe matchup-specific weaknesses, but it is not an equivalent reformulation of DRQ’s multiplayer fitness. The present results do not establish that either objective is intrinsically preferable.

The next step is to evolve matched populations under each objective and compare their convergence, diversity, and held-out generality. Further controlled experiments should also vary

opponent composition while examining battle dynamics to identify the multiplayer mechanisms that remain unexplained. Changing an evaluator can therefore alter what the system preserves and may redirect its evolutionary trajectory.

Use of language-model assistance

I used OpenAI’s ChatGPT and Codex extensively as research assistants throughout this project. They helped me study the supplied papers, blog posts, and codebases; explore and progressively refine the research question; design the experiments and evaluation metrics; implement and debug analysis code; inspect execution logs and experimental outputs; generate figures; and draft and revise this manuscript. In particular, the pairwise formulation and the sequence of diagnostic experiments were developed iteratively through discussions with these tools.

The models were used to accelerate exploration and analysis, not as sources of experimental evidence. Their suggestions were treated as provisional and were sometimes revised or discarded after closer inspection. I checked the resulting claims against the original papers, source code, commit history, execution logs, and saved simulator outputs. I made the final decisions regarding the research question, experimental design, interpretations, and manuscript structure, and personally executed the experiments and reviewed the final text. All quantitative results and figures were computed from the saved experimental data.

References

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